

Microprocessors and Microcontrollers

Assignment 7- Week 7

TYPE OF QUESTION: MCQ

Number of questions: 15

Total mark: 15 X 1 = 15 M

QUESTION 1:

Correct Answer: (c)

Detailed Solution:

The **TMOD (Timer Mode) register** in **8051 microcontrollers** is an **8-bit register** used to configure the two timers (**Timer 0 and Timer 1**). It is structured as follows:

Bit	7	6	5	4	3	2	1	0
Function	Gate	C/T	M1	M0	Gate	C/T	M1	M0
Timer 1	X	X	M1	M0	-	-	-	-
Timer 0	-	-	-	-	X	X	X	X

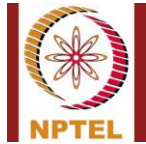
- Lower 4 bits configure Timer 0.
- Upper 4 bits configure Timer 1.

Decoding the Given Constraints

1. **Program Timer 1** → We focus on the **upper 4 bits** of TMOD.
2. **Mode 1** → Timer mode **M1M0 = 01** (16-bit Timer Mode).
3. **Use 8051 XTAL for the clock source** → Internal clock (**C/T = 0**).
4. **Start/Stop using internal instructions** → **Gate = 0** (controlled by TR1, not external hardware).

Setting Up TMOD for Timer 1

- **M1M0 = 01** (Mode 1: 16-bit timer mode)
- **C/T = 0** (Timer mode, not counter mode)
- **Gate = 0** (Start/Stop controlled by software)



Bit	7	6	5	4	3	2	1	0
Value	0	0	0	1	-	-	-	-

This gives 0001 0000 (10h).

QUESTION 2:

Correct Answer: (a)

Detailed Solution:

TR1 (Timer 1 Run Control bit) → Correct Answer

- This bit starts or stops Timer 1.
- TR1 = 1 → Timer 1 starts running.
- TR1 = 0 → Timer 1 stops.
- Controlled by software (via instructions like SETB TR1 or CLR TR1).

TF1 (Timer 1 Overflow Flag) → Incorrect

- This flag is set when Timer 1 overflows.
- It does not control the start/stop of Timer 1 but indicates overflow.

IE1 (External Interrupt 1 Edge Flag) → Incorrect

- This is related to external interrupts and has nothing to do with Timer 1.

IT1 (Interrupt Trigger Type for INT1) → Incorrect

- This bit determines whether INT1 (External Interrupt 1) is edge- or level-triggered.
- It does not control Timer 1.

QUESTION 3:

Correct Answer: c

Detailed Solution:



SBUF (Serial Buffer Register) → Incorrect

- This register is used for serial data transmission and reception, but it does not contain SMOD.

TMOD (Timer Mode Register) → Incorrect

- This register is used to configure timers (Timer 0 and Timer 1), but not related to serial communication or SMOD.

PCON (Power Control Register) → Correct Answer

- PCON Register contains the SMOD bit (Bit 7).
- SMOD = 1 → Doubles the baud rate in serial communication.
- SMOD = 0 → Normal baud rate.

TCON (Timer Control Register) → Incorrect

- This register controls timers and external interrupts but does not include SMOD.

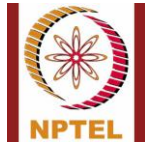
QUESTION 4:

Correct Answer: b

Detailed Solution:

Step 1: Understanding Timer Mode 1

- **Mode 1** is a **16-bit timer** mode.
- The timer counts from **TH1 TL1** to **FFFFh (65535 in decimal)** before overflowing.



Step 2: Calculate Initial Timer Value

Given:

- TH = 20h, TL = 0Fh
- Convert to decimal:

$$\begin{aligned}\text{Initial Value} &= 20h0Fh = (20h \times 256) + 0Fh \\ &= (32 \times 256) + 15 = 8192 + 15 = 8207\end{aligned}$$

- Timer counts from 8207 to 65535, so the total counts required:

$$\text{Total Counts} = 65536 - 8207 = 57329$$

Step 3: Calculate Time for One Count

- Crystal frequency = 11.0592 MHz
- 8051 Timer clock frequency:

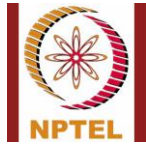
$$\text{Timer Clock} = \frac{\text{Crystal Frequency}}{12} = \frac{11.0592 \text{ MHz}}{12} = 921.6 \text{ kHz}$$

- Time per timer count:

$$T = \frac{1}{921600} = 1.085 \mu s$$

Step 4: Calculate Total Delay

$$\begin{aligned}\text{Delay} &= \text{Total Counts} \times T \\ &= 57329 \times 1.085 \mu s \\ &= 62.201 \text{ milliseconds}\end{aligned}$$



QUESTION 5:

Correct Answer: a

Detailed Solution:

Mode	Bits Used	Description
Mode 0	13-bit timer	8-bit THx + 5-bit TLx (Lower 5 bits of TL)
Mode 1	16-bit timer	Full 16-bit timer (THx and TLx)
Mode 2	8-bit auto-reload	TLx acts as counter, THx holds reload value
Mode 3	Split mode	Two 8-bit timers (Timer 0 only)

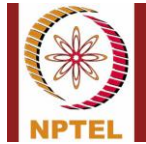
- Mode 0 (13-bit timer) counts from 0000h to 1FFFh (8192 counts).
- Used in older 8051 designs but rarely preferred today.

QUESTION 6:

Correct Answer: a

Detailed Solution:

The serial communication module (UART) in 8051 is controlled by the SCON (Serial Control) register. P3.0 (RxD) → Used to receive data (serial input). P3.1 (TxD) → Used to transmit data (serial output). These pins are part of Port 3, which has dedicated alternate functions.



QUESTION 7:

Correct Answer: a

Detailed Solution:

The default interrupt priority order in the 8051 microcontroller (from highest to lowest priority) is:

1. INT0 (External Interrupt 0)
2. TF0 (Timer 0 Overflow Interrupt)
3. INT1 (External Interrupt 1)
4. TF1 (Timer 1 Overflow Interrupt)
5. RI + TI (Serial Communication Interrupt)

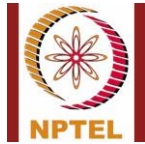
Interrupt	Interrupt Source	Default Priority (High → Low)
INT0	External Interrupt 0	1st (Highest Priority)
TF0	Timer 0 Overflow	2nd
INT1	External Interrupt 1	3rd
TF1	Timer 1 Overflow	4th
RI + TI	Serial Communication	5th (Lowest Priority)

- If two interrupts occur simultaneously, the higher priority one gets serviced first.
- The priority order can be modified using the IP (Interrupt Priority) register, but by default, it follows the above order.

QUESTION 8:

Correct Answer: (b)

Detailed Solution:



Step 1: Understanding the IP (Interrupt Priority) Register

The 8051 Interrupt Priority (IP) register assigns high priority to interrupts when a bit is set to '1'. The bit positions in the IP register are:

Bit	7	6	5	4	3	2	1	0
Interrupt	-	-	-	Serial (TI/RI)	TF1 (Timer 1)	INT1 (External Interrupt 1)	TF0 (Timer 0)	INT0 (External Interrupt 0)

- TF0 (Timer 0 Overflow Interrupt) corresponds to Bit 1.
- To assign high priority to TF0, Bit 1 must be set to '1'.

Step 2: Identifying the Correct Instruction

- Binary for setting Bit 1 (TF0) high:

$$00000010B = 02h$$

- The instruction that sets this correctly is:

MOV IP, #00000010B

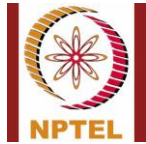
Step 3: Verifying Answer Choices

Option	Instruction	Binary Value	Effect	Correct?
(a)	MOV IP, #00000100B	00000100B (04h)	Assigns high priority to INT1, not TF0	✗ Incorrect
(b)	MOV IP, #00000010B	00000010B (02h)	✓ Assigns high priority to TF0	✓ Correct
(c)	MOV IP, 08h	00001000B	Assigns high priority to Serial Interrupt	✗ Incorrect
(d)	MOV IP, #01h	00000001B	Assigns high priority to INT0	✗ Incorrect

QUESTION 9:

Correct Answer: a

Detailed Solution:



Explanation:

QUESTION 10:

Correct Answer: C

Detailed Solution:

Step 1: Understanding the IP (Interrupt Priority) Register

The Interrupt Priority (IP) register in the 8051 microcontroller determines the priority of interrupts.

Bit	7	6	5	4	3 (TF1)	2 (INT1)	1 (TF0)	0 (INT0)
Interrupt	-	-	-	Serial (RI/TI)	TF1	INT1	TF0	INT0

- If a bit is set to '1', the corresponding interrupt is assigned **higher priority**.
 - If multiple interrupts have the same priority, the **default priority order** applies.
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Step 2: Analyzing the Given Instruction

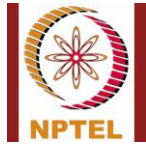
The instruction provided is:

MOV IP, #00001100B

- **00001100B** in binary means:
 - Bit 3 (TF1) = 1 → High priority
 - Bit 2 (INT1) = 1 → High priority
 - Bits 1 & 0 (TF0 and INT0) = 0 → Default priority

So, after this instruction:

1. TF1 and INT1 are given high priority.
2. TF0 and INT0 remain at default priority.
3. RI + TI (Serial Interrupt) remains lowest priority.



Step 3: Determining the Correct Order

Since TF1 and INT1 are high-priority interrupts, we compare them using the default order:

- INT1 has higher priority than TF1 (as per the default order).
- TF1 comes after INT1 (since both are high priority).
- INT0 and TF0 remain in their default order ($INT0 > TF0$).
- RI + TI is the lowest.

Thus, the new priority order after updating IP is:

✓ INT1 > TF1 > INT0 > TF0 > RI+TI

QUESTION 11:

In idle mode of power control register, which of the following is not true?

- a) All of registers, ports and internal RAM maintain their data
- b) The ALE and PSEN output are held low
- c) The internal CPU clock is gated off
- d) Interrupt, Timer and Serial port functions act normally

Correct Answer: b

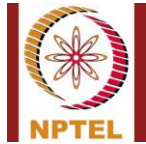
Detailed Solution:

Step 1: Understanding Idle Mode in 8051 Microcontroller

The 8051 microcontroller has two power-saving modes:

1. Idle Mode
2. Power-Down Mode

In Idle Mode, the CPU is stopped to save power, but peripherals like timers, interrupts, and the serial port continue functioning.



Step 2: Characteristics of Idle Mode

When the 8051 enters Idle Mode:

- Registers, ports, and internal RAM retain their data (Correct).
- ALE (Address Latch Enable) and PSEN (Program Store Enable) continue operating normally. (Incorrect statement)
- The internal CPU clock is gated off (stopped) (Correct).
- Interrupt, Timer, and Serial Port functions still operate (Correct).

QUESTION 12:

In serial communication which of the following mode of operation receives or transmits 10 bits

- a) Mode 0
- b) Mode 1
- c) Mode 2
- d) Mode 3

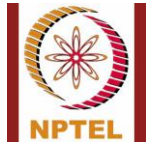
Correct Answer: b

Detailed Solution:

Step 1: Understanding Serial Communication Modes in 8051

The 8051 microcontroller **has** four serial communication modes **for transmitting and receiving data using the** Serial Control (SCON) register.

Mode	Bits Transmitted/Received	Baud Rate	Start & Stop Bits
Mode 0	8 bits	Fixed	No start/stop bits
Mode 1	10 bits (8 data + Start + Stop bit)	Variable	✔ Uses Start & Stop bits
Mode 2	11 bits (9 data + Start + Stop bit)	Fixed	Yes
Mode 3	11 bits (9 data + Start + Stop bit)	Variable	Yes



Step 2: Identifying the Correct Answer

- The question asks for a mode that transmits/receives 10 bits.
- Mode 1 transmits 8 data bits + 1 Start bit + 1 Stop bit = 10 bits.
- Other modes do not match exactly:
 - Mode 0 → 8 bits (No start/stop bits)
 - Mode 2 & Mode 3 → 11 bits (9 data + start + stop)

QUESTION 13:

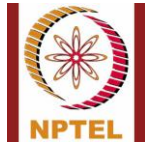
The following program generates a square wave on pin P1.5 Using timer1, Find the frequency.
Consider XTAL = 11.0592 MHz.

```
MOV TMOD, #10H  
  
AGAIN: MOV TL1, #34H  
  
MOV TH1, #12H  
  
SETB TR1  
  
BACK: JNB TF1, BACK  
  
CLR TR1  
  
CPL P1.5  
  
CLR TF1  
  
SJMP AGAIN
```

- a) 3.25 Hz
- b) 15.14 Hz
- c) 7.57 Hz
- d) 6.57 Hz

Correct Answer: c

Detailed Solution:



Step 1: Understanding the Timer Mode

- The instruction `MOV TMOD, #10H` sets **Timer 1 in Mode 1 (16-bit timer)**.
- `TL1 = 34H`, `TH1 = 12H` → The timer is loaded with **1234H**.

Since **Timer 1** is a **16-bit timer**, it counts from **1234H** to **FFFFH** before rolling over.

Step 2: Calculate Timer Overflow Time

The 8051 system clock (XTAL) is 11.0592 MHz, but the timer clock is:

$$\text{Timer Clock} = \frac{\text{XTAL}}{12} = \frac{11.0592 \text{ MHz}}{12} = 921.6 \text{ kHz}$$

So, each timer tick takes:

$$\frac{1}{921600} \approx 1.085 \mu\text{s}$$

Now, calculating the initial timer value:

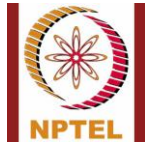
$$\text{Initial Value} = 1234H = 4660 \text{ (decimal)}$$

Since the timer overflows at `FFFFH` = 65535,

$$\text{Counts to Overflow} = 65535 - 4660 = 60875$$

The time taken for one overflow:

$$\begin{aligned} \text{Overflow Time} &= 60875 \times 1.085 \mu\text{s} \\ &= 66000 \mu\text{s} = 66.0 \text{ ms} \end{aligned}$$



Step 3: Determine the Frequency

- The CPL P1.5 toggles P1.5 once per overflow, meaning one full cycle (high + low) takes:

$$2 \times 66.0 \text{ ms} = 132 \text{ ms}$$

The frequency of the square wave is:

$$f = \frac{1}{T} = \frac{1}{132 \times 10^{-3}} \\ = 7.57 \text{ Hz}$$

QUESTION 14:

Find the value for TMOD to program timer 1 in mode 1 and use external clock source.

- a) TMOD = 1111 0110
- b) TMOD = 0101 0000
- c) TMOD = 1111 0101
- d) TMOD = 1010 0000

Correct Answer: b

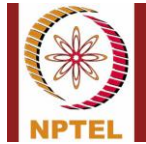
Detailed Solution:

The TMOD (Timer Mode) register is an 8-bit register that configures Timer 0 and Timer 1 in the 8051 microcontroller.

TMOD Format:

$$\text{TMOD} = [\text{GATE1 C/T1 M1 M0}] [\text{GATE0 C/T0 M1 M0}]$$

- Upper 4 bits (Timer 1)
- Lower 4 bits (Timer 0)



Step 1: Setting Timer 1 in Mode 1

- Mode 1: 16-bit timer ($M1 = 0, M0 = 1$)
 - $M1 = 0, M0 = 1$

Step 2: Using External Clock Source

- $C/T = 1$ (Counter mode, external clock from T1 pin)
- $GATE = 0$ (Timer enabled by TR1 only, not controlled by INT1)

Thus, for Timer 1 (Upper 4 Bits):

$$GATE1 = 0, C/T1 = 1, M1 = 0, M0 = 1$$

0101

Step 3: Keeping Timer 0 Unchanged

- Since the question only asks about Timer 1, the lower 4 bits (Timer 0 settings) can be 0000.

Thus, $TMOD = 0101\ 0000$ (0x50).

Step 4: Verifying Answer Choices

Option	TMOD Value (Binary)	Meaning
(a)	1111 0110	Incorrect (Mode 2 used, Timer 0 affected) ✗
(b)	0101 0000	Correct (Timer 1 in Mode 1 with External Clock) ✓
(c)	1111 0101	Incorrect (Affects both timers, not required) ✗
(d)	1010 0000	Incorrect (Wrong mode selection) ✗

Refer lecture slides

*****END*****